

Data Driven Approach for Reduced Value at Risk Forecasts in Renewable Power Supply Systems

Behrouz Banitalebi
Department of Supply Chain
Management
University of Manitoba
Winnipeg, Canada
banitalb@myumanitoba.ca

Srimantoora S. Appadoo
Department of Supply Chain
Management
University of Manitoba
Winnipeg, Canada
appadoo@cc.umanitoba.ca

Aerambamoorthy Thavaneswaran
Department of
Statistics
University of Manitoba
Winnipeg, Canada
thavane@umanitoba.ca

Abstract— Electricity production from renewable resources such as wind and solar has increased uncertainty in the electricity supply chains. This uncertainty fluctuates the electricity price and consequently causes a highly volatile electricity market and increases the value at risk (VaR) of electricity price. Therefore, production managers need to have an accurate forecast of electricity price as well as VaR before making any plan for further production. In this work, performance of exponentially weighted moving average (EWMA) and recently introduced generalized-EWMA (G-EWMA) are evaluated for VaR forecasting. Both methods have been applied to electricity price dataset of different Canadian provinces. The results of our data analysis show that G-EWMA perform more accurately than EWMA. In addition, we show that Ontario has the highest electricity VaR among other Canadian provinces and this would be the consequence of participation of wind and solar power plants in electricity production system. The electricity market, which is responsible for scheduling electricity buyers and sellers, should also use forecasting tools for matching supply and demand to avoid any sudden change in electricity price. Double exponential smoothing (DES) and triple exponential smoothing (TES) forecasting methods have been used for electricity supply and demand forecast. Our analyses show that DES forecasts of supply/demand outperform TES.

Keywords— *Electricity market, supply, demand, price, volatility, VaR, EWMA, G-EWMA, DES, TES, Backtesting.*

I. INTRODUCTION

In the centralized electricity supply chains, the electricity production process is managed by a single power plant with no competitor. As a result, the electricity producer has authority to set electricity prices. Technology development in power engineering section has made it possible for investors to implement micro-grids and to start producing electricity at both large and small scale. Deployment of micro-grids has transformed the structure of electricity production to a decentralized system in which they can freely participate in electricity market and offer their own price and capacity for electricity. The price of wholesale electricity is changing every day or even every hour and hence managing the market has become more difficult. In fact, improper management of decentralized electricity supply chain causes high variation in electricity price and consequently affects other commodity prices.

Electricity market can be highly affected by dramatic fluctuations of supply and demand. Therefore there should be an accurate way to assess risk in electricity production. VaR is a risk evaluation tool that quantifies the level of possible loss for an investment during a period of time. Statistical processes such as the generalized autoregressive conditional heteroskedasticity

(GARCH) model have been applied in this context to forecast VaR in energy market. For example, the performance of fractionally integrated GARCH (FIGARCH) hyperbolic GARCH (HYGARCH) was evaluated on electricity market VaR forecasting in [1] where a normal and t-distribution were considered. Under these conditions, the FIGARCH model provides a more accurate forecast of VaR. A bivariate GARCH-in-mean was used in [2] to calculate VAR and to investigate its impact on returns under high and low uncertainty situations in energy market.

Other statistical methods have been used in order to find a more accurate forecast of VaR. For instance, an empirical mode decomposition (EMD) algorithm, combined with copula theory as in [3], was used to construct the EMD-Copula model to obtain VaR forecast of the electricity price. Moreover, the proposed method in [3] outperforms the dynamic conditional correlation GARCH model in VaR forecasting. Furthermore, GARCH and EWMA models have been applied in scenarios with different level of volatility for VaR forecasting in [4]. It was concluded that EWMA VaR forecasts are more reliable than GARCH. Recently, some studies [5-7] have considered fuzzy approaches to increase the accuracy of risk forecasting. For instance, the authors in [5] introduced a direct data-driven predictive model, which employs neural networks, to forecast conditional volatility.

Supply and demand forecasting is an effective way to minimize uncertainty in electricity market and eventually to reduce price volatility. In more details, a more accurate forecast of one day ahead suppliers' production and consumers' consumption would allow better scheduling of producers and smoother variation of electricity price. Moreover, various forecasting methods have been considered, upscaling models in [8], artificial neural networks in [9], Markov chains in [10], ARMA and support vector machine in [11], Monte Carlo simulation, and general algebraic modeling system model in [12] and [13]. Electricity produced from solar and wind, as the main source of uncertainties in electricity market, have drawn researchers' attention. As a result, stochastic forecast methods have been used to improve prediction of supply/demand. For instance, vector autoregression is employed for wind forecasting in [14], with lasso estimation being used to avoid overfitting. Stochastic optimization in [15] is used to propose a decision making framework for wind power plants to be able to participate in electricity trading more confidently. Machine learning methods have been used widely in order to decrease the forecast error of renewable power plants production [16]-[20].

In this paper, we compared accuracy of EWMA with a new forecast model namely Generalized EWMA (G-EWMA) which

was proposed in [8] for Canadian electricity market VaR forecasting. Through empirical analysis and simulation we show that why G-EWMA outperformed than EWMA. In addition we have extended contribution of forecasting tools in electricity market by employing Double Exponential Smoothing (DES) as well as Triple Exponential Smoothing (TES) for demand and supply prediction. All the empirical analysis and simulation have been done using MATLAB and all the codes are available on <https://github.com/Bani-Talebi/Electricity-Price-Value-at-Risk-VaR->.

The rest of this paper is organized as: electricity supply chain is described in Section II, basics of EWMA and G-EWMA for VaR forecasting are introduced in Section III, and DES and TES are used to obtain supply/demand forecasts in Section IV. A dataset of Canadian electricity statistics is used to analyze the performance of mentioned methods in both Sections III and IV.

II. ELECTRICITY SUPPLY CHAIN

Electricity supply chain consists of power generation plants, electricity markets (which can be divided in wholesale and retail markets), transmission lines, distributors, and final costumers. In fact, production, transmission and distribution are three main functions of this structure, and they exist in both centralized and decentralized electricity supply chains. Electricity markets are responsible for regulatory functions, and micro-grids can be considered as self-producers and in some cases may sell their extra electricity to the other customers.

A. Electricity Production

In a power plant, generators produce electricity using energy resources which can determine the type of power plant in an area. For example, in Canada hydroelectric plants are widely used in Manitoba, Quebec and BC due to the abundance of water resources. Thermal power plants are another class that have been mainly established in Alberta, Saskatchewan and Ontario. In thermal plants, the turbines are exposed to high pressure steam in order to make mechanical energy which is finally transformed to electricity using a generator. The required energy for producing high pressure steam can be obtained by burning fossil fuels or splitting uranium atoms in a nuclear fission process. In Saskatchewan, which has an abundance of coal, and Alberta, which is rich in oil and natural gas, the combustion thermal plants are more common, while Ontario has made significant investments in nuclear energy. Renewable power plants are also growing slowly in Canada and getting more popular because of their clean energy production. These plants convert energy of renewable resources, such as wind or solar to electricity.

B. Main and Micro Grid

The main grid is the combination of transmission and distribution networks. The most important duty of the grid is to transfer high voltage electricity from producers to distribution centers and deliver it to final customers after reducing the voltage. The grid should have enough capacity to cover all sorts of electricity needs, and high capacity grids may have high complexity which can affect system performance. One solution for decreasing the complexity of grids is to establish small power station, on the supply side. Renewable energy resources, such as solar and wind, have provided opportunity for final customers to

install a small power generator, which can be referred to micro-grid, to meet their own need and also in some cases sell electricity to the main grid. Traditionally, electricity supply chains have had a centralized structure in which power plants were the only power generators. Nowadays the structure has been shifted to decentralized forms where micro-grids participate in power generation.

C. Electricity Market

In delivering electricity to the final customers, electricity markets are responsible for setting electricity price and also balancing electricity supply and demand. Because electricity is such an crucial commodity that needs to be available at all times, electricity markets have to predict the amount of consumption and schedule producers so that the supply meet the demand for the next 5 minutes, 1 hour or next day. They are also responsible for setting the electricity prices which depend on generation capacity and consumers' demand. In fact, each producer offers its predicted capacity as well as a price. In most cases hydroelectric has the cheapest price which is followed by nuclear power. Renewable power plants often have less capacity than other plants and they may offer higher price in the market because their equipment are more expensive than those used in traditional plants. However, in most cases electricity from fossil fuel plant has the highest price because of carbon tax charges. The electricity market gives priority to hydroelectric (because of its cheap price) and nuclear power plants (because of its fixed capacity). The highest offer will be the electricity price for the whole market.

In the centralized electricity supply chain, there is only one producer, and as a result the electricity market is mostly stable. But the expansion of micro-grids has provided an opportunity for many producers to compete in electricity markets which consequently increases volatility in electricity markets. In fact, the electricity market is more volatile than other commodity markets because electricity cannot be stored and has to be bought prior to consumption. On the other hand, both supply and demand are random variables that are hard to match. This can be seen in figure 1, where hourly data of electricity production, demand and price in Ontario is presented.

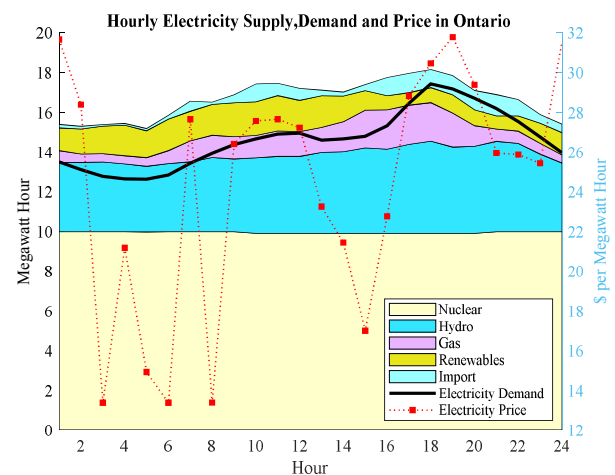


Fig. 1. Ontario Hourly Electricity data

Figure 1 shows the contribution of different power plants in Ontario electricity market. In all 24 hours, a fixed level of nuclear plants' output is reserved because in these type of plants the output cannot change rapidly to meet the sudden variation of customer demands. Hydroelectric power plants are more flexible than nuclear plants in terms of increasing or decreasing output and also can provide a fixed amount of electricity for all 24 hours. In contrast capacity of renewable power plants is not fixed because it mainly depends on 2 other random variables: wind and solar energy. Participation of these plants in the electricity market makes it harder to match supply and demand for the next day, hour or five minutes.

Figure 1 also indicates hourly electricity price in Ontario for the same day and shows how renewable power plants can change the wholesale price of electricity. For example, the price of electricity reaches its peak between 4:00 pm and 8:00 pm when all the producers are participating in the market. High variation of electricity price could increase risk for electricity producers because they may encounter negative prices in electricity markets. Therefore, an accurate forecast method is required.

III. VAR FORECAST OF ELECTRICITY PRICE

The set of daily electricity price, P_t , $t = 1, 2, \dots, n$, where t is the time index is a non-stationary time series. Similar to other stock price time series, P_t has geometric Brownian motion (GBM) which can be defined as

$$P_t = e^{\mu t + \sigma \omega_t} \quad (1)$$

where μ and σ denote mean and volatility of time series respectively and ω_t is a Wiener process or Brownian motion. On the other hand, log return of electricity price,

$$r_t = \log P_t - \log P_{t-1}, \quad (2)$$

can be considered as a stationary process. Forecast of VaR at time $t+1$ could be calculated as

$$VaR_{t+1}(p) = -\sigma_{t+1} F_r^{-1}(p), \quad (3)$$

where σ_{t+1} indicated volatility forecast and $F_r^{-1}(p)$ is the inverse Cumulative Distribution Function (CDF) of the observations at confidence level (p). Therefore, the first step for VaR forecasting, VaR_{t+1} , is to find σ_{t+1} . In this section σ_{t+1} is predicted by EWMA and G-EWMA and accuracy of these two forecasting methods is compared. As a matter of fact, variance of σ_{t+1} will have smaller value for the more accurate forecast method.

A. Volatility Forecasting with EWMA

The conventional way to find the forecast of volatility at time $t+1$, $\hat{\sigma}_{t+1}$, is by taking square root of the one step ahead forecast $\hat{\sigma}_{t+1}^2$ which can be predicted by EWMA. So, the initial value for volatility forecast using EWMA could be defined as

$$S_0 = \frac{1}{N-1} \sum_{t=1}^N (r_t - \bar{r})^2 \quad (4)$$

Where $\bar{r}$ is assumed to be mean of the log returns and N is the number of log returns. Consequently, according to [20] the variance can be derived as

$$S_t = (1 - \alpha)(r_t - \bar{r})^2 + \alpha S_{t-1}. \quad (5)$$

In this equation α is the smoothing constant between 0 and 1 and can be obtained by minimizing the one step ahead forecast error sum of squares that is minimizing the following equation

$$\min_{\alpha} \sum_{t=1}^N [(r_t - \bar{r})^2 - S_{t-1}]^2. \quad (6)$$

Finally, the volatility forecast can be obtained as

$$\hat{\sigma}_{t+1} = \sqrt{S_t}. \quad (7)$$

B. Volatility Forecasting with G-EWMA

G-EWMA has been proposed by Thavaneswaran *et al* in [5] to directly forecast the volatility instead of variance. In this method,

$$\hat{\sigma}_{t+1} = (1 - \alpha)\hat{\sigma}_t + \alpha \frac{|r_t - \bar{r}|}{\rho} \quad (8)$$

where

$$\rho = \text{Corr}(r_t - \bar{r}_t, \text{sgn}(r_t - \bar{r}_t)) \quad (9)$$

and Corr. is the correlation function.

It can be proved that for any $0 < \rho < 0.79$ the degree of freedom (ν) for a t-distribution obtained by solving the nonlinear equation

$$2\sqrt{\nu - 2} = \rho(\nu - 1) \text{Beta} \left[\frac{\nu}{2}, \frac{1}{2} \right] \quad (10)$$

and for $\rho > 0.79$ the normal distribution can be considered. It can be shown that the asymptotic variance of the sample standard deviation is

$$\text{As. Var}(s) = \frac{(k + 2)}{4} \frac{\sigma^2}{n}, \quad (11)$$

in which k is excess kurtosis. This equation indicates that asymptotic variance depends on kurtosis that can have very large value in some cases and results in wrong volatility estimation. Moreover, it can be proved that asymptotic variance of G-EWMA estimator can be calculated as

$$\text{As. Var}(\hat{\sigma}) = \frac{1}{n} \left(\frac{1 - \rho^2}{\rho^2} \right) \sigma^2. \quad (12)$$

According to this equation, the asymptotic variance in this method does not depend on kurtosis which makes this forecast very useful when degrees of freedom is less than 4. In more details, kurtosis of t distribution has very large values when $\nu < 4$ and consequently EWMA as shown in Figure 2 has a poor performance. In fact, the two curve will merge for large ν which means that the two forecast methods will have the same results for large degrees of freedom.

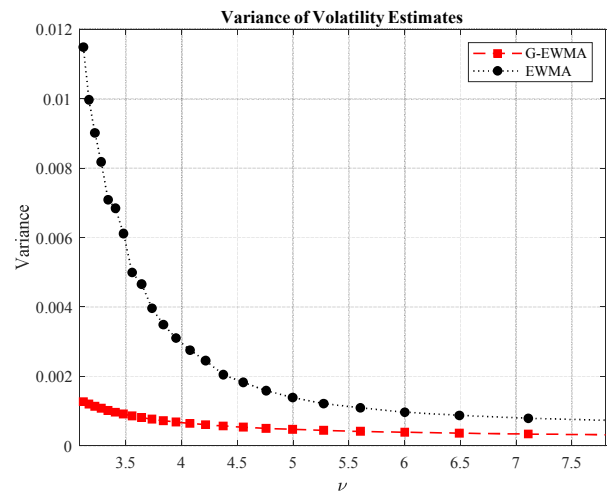


Fig. 2. Variance of volatility forecast for different degrees of freedom

However, the results of our analysis show that for all daily electricity price data sets in Canadian provinces, the degrees of freedom for the t-distribution vary between 3.5 and 4.00 which means that there is a significant difference between performance of EWMA and G-EWMA in volatility forecasting. For this reason, G-EWMA has been used in this paper to forecast electricity VaR.

C. Results and Analysis

G-EWMA is applied to whole sale electricity price in most populated Canadian provinces such as Ontario, Quebec, BC, Alberta, Manitoba and Saskatchewan. Table 1 presents participation of different power resources in electricity production of Canadian provinces. According to this table, Ontario has the most diverse electricity market because all four types of power resources have been used in this province while other provinces mainly rely on one type of power generation. For example, the majority of electricity production in Quebec, Manitoba and BC comes from hydroelectric power plants. And, in provinces of Alberta and Saskatchewan the fossil fuel power plants are the dominant ones.

TABLE 1. ELECTRIC POWER SUPPLY BY TYPE OF ELECTRICITY RESOURCES FOR DIFFERENT PROVINCES IN AUGUST 2019.

Power Resources	Percentage of Produced Electricity in Provinces					
	ON	QC	MB	SK	AB	BC
Hydro Electric	19.50	92.6	97.65	15.60	4.23	84.84
Nuclear	63.81	0	0	0	0	0
Fossil Fuels	10.50	0.94	0.12	81.15	92.43	12.91
Wind power	3.90	6.45	0	3.25	3.28	2.24
Solar	2.29	0.01	2.23	0	0.06	0.01

Source: Statistics Canada. Table 25-10-0015-01 Electric power generation, monthly generation by type of electricity DOI: <https://doi.org/10.25318/2510001501-eng>

Table 2 shows the VaR forecast values of monthly electricity price in different Canadian provinces. The G-EWMA method has been applied to monthly electricity price since Jan. 1981 to forecast electricity VaR for 466th month. The tail property (p) has been set to 0.01 which means that the amount of loss in monthly electricity price, 1 in 100 month, would be greater than the VaR forecast values. As expected, Ontario has the highest VaR and the reason might be the penetration of both solar and wind power plants in electricity market in this province. In addition, the electricity price VaR in Alberta is relatively high compared to Quebec, BC, Manitoba and Saskatchewan, because more than 92 percent of electricity in this province is produced by burning oil and gas and variation of oil and gas price will have a direct impact on electricity market.

TABLE 2. VAR IN MONTHLY ELECTRICITY PRICE OF CANADIAN PROVINCES (CAD/MWH)

$p=0.01$	AB	BC	MB	ON	QC	SK
VaR_{466}	7.60	0.49	0.48	11.70	0.43	0.48

Figure 3 shows the results of back testing for the two VaR forecasts with confidence level 0.01. It proves the validity of G-EWMA performance in which the amount of lost is greater than predicted VaR. Although this point is predicted by EWMA as well, the result of this forecast tool is not as accurate as G-EWMA. Because the graph shows that there is another VaR at time index=410 which is not a valid prediction.

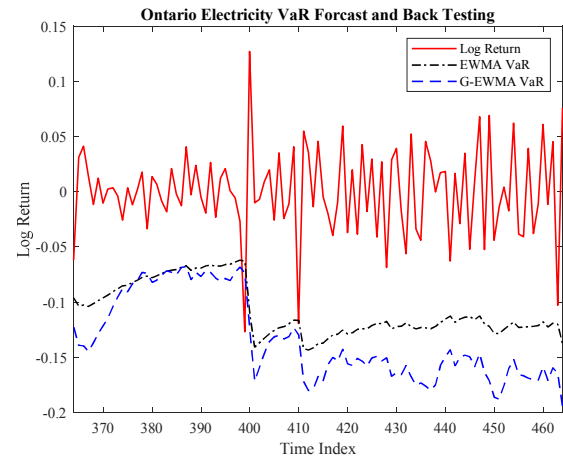


Fig. 3. Backtesting for last 100 log returns of Ontario monthly electricity price

IV. ELECTRICITY SUPPLY AND DEMAND FORECASTING

As mentioned earlier having uncertainty in electricity supply and demand will result in a high volatility electricity market. Therefore, supply and demand forecasting tools should be applied in order to reduce uncertainty from electricity supply and demand which eventually alleviates the price fluctuations. Figure 4 shows that how electricity price is set in a day ahead market where sellers and buyers are participating in an auction. The figure indicates the fact that consumers' bids are ranked in a decreasing price order while suppliers' offers are ranked in an increasing way which is called the merit order. The point where two graphs cross, equilibrium point, is determined the market price of electricity for the next day. An accurate forecast of demand and supply will help electricity market to schedule suppliers properly in a way that they can meet the whole demand for next day. Otherwise, there is a need for imported electricity which has a high price and will increase the VaR of this commodity. In this section, DES and TES which can be used for non-stationary data such as electricity supply and demand are briefly described according to [20].

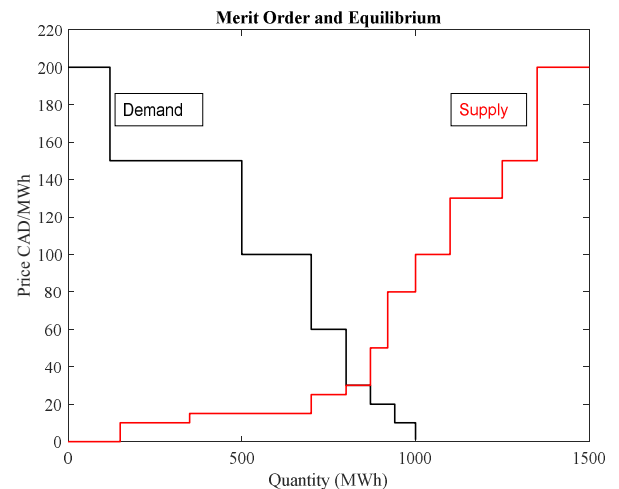


Fig. 4. Merit order and equilibrium

A. Double Exponential Smoothing

Similar to the electricity price time series, electricity supply and demand might have a non-stationary behavior. Therefore, EWMA is not able to find a proper forecast for supply and demand prediction and a more capable method is needed. DES can be applied to non-stationary data sets as long as there exist a linear regression model for them as

$$d(t) = \beta_0 + \beta_1 t + \varepsilon_t, \quad (13)$$

where $d(t)$ denotes electricity demand (supply, $s(t)$) and ε_t is a white noise. DES has 3 main steps to forecast 1 step ahead of d_n which is shown by $\hat{d}_n$. Step 1: Initialization

$$T_0^{(1)} = \beta_0 - \frac{1-\alpha}{\alpha} \beta_1, \quad (14)$$

$$T_0^{(2)} = \beta_0 - 2 \frac{1-\alpha}{\alpha} \beta_1$$

Step 2: Forecasting 1-step ahead

$$\hat{d}_n = \left(2 + \frac{\alpha}{1-\alpha}\right) T_n^{(1)} - \left(1 + \frac{\alpha}{1-\alpha}\right) T_n^{(2)} \quad (15)$$

Step 3: Updating

$$\begin{aligned} T_n^{(1)} &= \alpha d_n + (1-\alpha) T_{n-1}^{(1)} \\ T_n^{(2)} &= \alpha T_n^{(1)} + (1-\alpha) T_{n-1}^{(2)} \end{aligned} \quad (16)$$

The optimal smoothing constant α can be obtained by minimizing the one step ahead forecast error sum of squares

$$SSE(\alpha) = \sum (d_t - (2 + \frac{\alpha}{1-\alpha}) T_{t-1}^{(1)} + (1 + \frac{\alpha}{1-\alpha}) T_{t-1}^{(2)})^2. \quad (17)$$

B. Triple Exponential Smoothing

The idea behind TES is almost the same at DES concept but instead of 2 smoothing steps, there will be 3 steps and the reason arise from the fact that data are fitted by a constant quadratic model as

$$d(t) = \beta_0 + \beta_1 t + \beta_2 \frac{t^2}{2} + \varepsilon_t. \quad (18)$$

Using this model and assuming $\omega = 1 - \alpha$, the initial values for the smoothed statistic can be express as

$$\begin{aligned} T_0^{(1)} &= \beta_0 - \frac{\omega}{\alpha} \beta_1 + \frac{\omega(1-\omega)}{2\alpha^2} \beta_2 \\ T_0^{(2)} &= \beta_0 - \frac{2\omega}{\alpha} \beta_1 + \frac{2\omega(1-2\omega)}{2\alpha^2} \beta_2 \\ T_0^{(3)} &= \beta_0 - \frac{3\omega}{\alpha} \beta_1 + \frac{3\omega(1-3\omega)}{2\alpha^2} \beta_2 \end{aligned} \quad (19)$$

The 1-step ahead forecast can be calculated as

$$\begin{aligned} \hat{d}_n &= \left[3 + \frac{\alpha(1-5\omega)}{2\omega^2} + \frac{\alpha^2}{\omega^2}\right] T_n^{(1)} \\ &\quad - \left[3 + \frac{\alpha(1-4\omega)}{\omega^2} + \frac{\alpha^2}{\omega^2}\right] T_n^{(2)} \\ &\quad + \left[1 + \frac{\alpha(1-3\omega)}{2\omega^2} + \frac{\alpha^2}{2\omega^2}\right] T_n^{(3)} \end{aligned} \quad (20)$$

And smoothed statistics will be updated through following equations:

$$\begin{aligned} T_n^{(1)} &= (1-\omega) d_n + \omega T_{n-1}^{(1)} \\ T_n^{(2)} &= (1-\omega) T_n^{(1)} + \omega T_{n-1}^{(2)} \\ T_n^{(3)} &= (1-\omega) T_n^{(2)} + \omega T_{n-1}^{(3)} \end{aligned} \quad (21)$$

The optimum α which leads to the forecast with the least error will be obtained by minimizing

$$SSE(\alpha) = \sum (d_t - \hat{d}_t)^2. \quad (22)$$

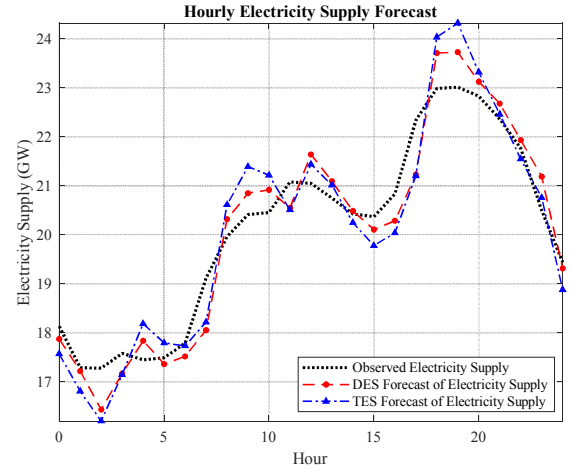


Fig. 5. Hourly electricity supply forecasts in Ontario for December 18, 2019

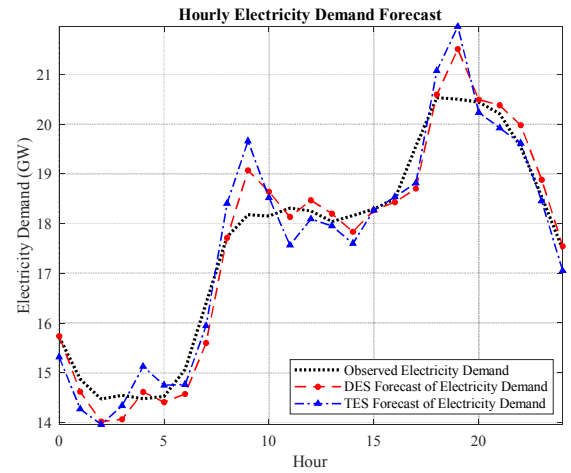


Fig. 6. Hourly demand forecasts of electricity in Ontario, December 18, 2019

C. Results and Analysis

Performance of DES and TES in hourly Ontario electricity supply and demand forecasting is compared in figures 5 and 6. The hourly electricity supply and demand in Ontario from Jan., 01, 2019 till Dec., 18, 2019 (which in total includes 8448 observation) is used and the results for last 24 hour is plotted in these two figures. In both cases DES has performed more accurate than TES, however it does not mean that in all cases DES works better than TES. Furthermore, these two methods can be implemented for wind and solar power energy to decrease the uncertainty of renewable power plants which seems to be a reason for high VaR in Ontario electricity market.

V. CONCLUSION

Participation of renewable resources in electricity production has decentralized the structure of the electricity supply chain. In the new architecture, renewable resources can be used for large or small scale power production. As a matter of fact, capacity of renewable power plants highly depends on stochastic parameters such as the availability and intensity of wind or solar energy. Therefore, the deployment of renewable power plants increases uncertainty in electricity supply and

demand, which consequently increases volatility in electricity price. In a high volatility electricity market VaR is a proper measure to determine the right strategy for power production. This paper has compared two forecast tools namely EWMA and G-EWMA, to find an accurate way to forecast electricity price VaR. In addition, we used G-EWMA to forecast electricity price VaR in Canadian provinces. Our analysis reveal that Ontario has the highest electricity price VaR among other provinces, which this may be a result of the uncertainty in solar and wind power plants in this province. Our analysis also concluded that Alberta has the second highest electricity price VaR, and other provinces are almost in the same range. Forecasting tools can be also employed by electricity markets for supply and demand forecasting. In fact, a proper scheduling of electricity buyers and sellers, which needs accurate supply/demand forecasts, would result in a stable market. In this work, DES and TES have been applied to Ontario hourly electricity supply/demand dataset, and the results of our analysis show that although both methods predicted supply and demand properly, DES was more accurate.

Future studies can be conducted to find out the main source of uncertainty with the highest impact on increasing electricity price VaR in Ontario. At first glance, this may conclude that renewable power plants have increased uncertainty in power supply/demand which has resulted in high volatility electricity prices. But this statement is not true in the province of Quebec, which has the highest power production from renewable resources and a very low electricity price VaR. There is also potential for evaluating two scenarios in which renewable-based micro-grids can choose to operate off-grid or semi off-grid. Semi off-grid is the situation where micro-grids mainly rely on their production and sell their surplus electricity to the main grid. But they are still allowed to buy power from the electricity market. Indeed, this scenario imposes uncertainty to electricity supply and demand. On the other hand, the uncertainty of off-grid renewable power plants has no impact on electricity price fluctuation. It might seem impractical to separate wind and solar power plants from the grid. In particular, for some of the European countries where renewables are the main source of power generation. But in some other places, such as Ontario, where the contribution of renewable resources to power generation is low, off-grid scenarios will help decrease electricity market VaR. The capacity of renewable micro-grids can be applied to other purposes such as electrolysis-based hydrogen production. The result can be used for fuel cell electric vehicles and can dramatically decrease the gas emission from transportation.

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